

1. Project Title

Retail Sales Data Engineering and Analytics Pipeline

2. Project Objective

The objective of this project is to design and develop an end-to-end data engineering solution using retail sales datasets. The project involves data ingestion, data cleaning, transformation, validation, exploratory data analysis, and dashboard development to generate meaningful business insights.

3. Dataset Description

Customers

Contains demographic information about customers.

- CustomerID: Unique identifier for each customer.
- FirstName, LastName: Customer's name.
- Gender: Customer's gender (M or F).
- BirthDate: Customer's date of birth.
- City: Customer's city of residence.
- JoinDate: Date the customer joined the loyalty program

Products

Contains details of all products sold.

- ProductID: Unique identifier for each product.
- ProductName: Name of the product.
- Category: Product category (e.g., Electronics, Fashion, Groceries).
- SubCategory: Product sub-category (e.g., Smartphone, Clothing, Dairy).
- UnitPrice: Selling price per unit.
- CostPrice: Cost per unit for the business.

Stores

Contains information about retail locations.

- StoreID: Unique identifier for each store.
- StoreName: Name of the store.

- City: City where the store is located.
- Region: Geographic region of the store (East, West, North, South).

Transactions

Records each individual sales transaction.

- TransactionID: Unique identifier for each transaction.
- Date: Date of the transaction.
- CustomerID: Foreign key linking to the Customers table.
- ProductID: Foreign key linking to the Products table.
- StoreID: Foreign key linking to the Stores table.
- Quantity: Number of units purchased.
- Discount: Discount rate applied to the transaction (e.g., 0.1 for 10%).
- PaymentMethod: Method of payment (e.g., Cash, Credit Card, Mobile Money).

4. Data Ingestion

All datasets were loaded from the source Excel workbook (retail_sales_dataset.xlsx) using the Python pandas library. Each of the four sheets (Customers, Products, Stores, Transactions) was imported into a separate pandas DataFrame for independent processing and subsequent merging.

5. ETL Pipeline Development

Extract

Data was extracted from the Customers, Products, Stores, and Transactions sheets.

Transform

The following transformations were performed:

- **Merging: The Transactions table was merged with the Products table using ProductID, the Customers table using CustomerID, and the Stores table using StoreID to create a unified analytical dataset. :**
 - Created CustomerName by concatenating FirstName and LastName.
 - Calculated Sales for each transaction using the formula:

$$\text{Sales} = \text{Quantity} \times \text{UnitPrice} \times (1 - \text{Discount})$$
 - Calculated Profit for each transaction using the formula:

$$\text{Profit} = \text{Sales} - (\text{Quantity} \times \text{CostPrice})$$

6. Data Validation and Quality Checks

The following data quality checks were performed on the final dataset to ensure its integrity for analysis:

Quality Check	Result	Description
Missing Values	0	No null values found in critical fields.
Duplicate Records	0	No duplicate transaction records found.
Invalid Quantity	0	All Quantity values are greater than 0.
Invalid Discount	0	All Discount values are between 0 and 1 (inclusive).
Negative Sales	0	Calculated Sales are non-negative.
Negative Profit	0	Calculated Profit is negative in some cases, indicating a loss-making sale (e.g., due to high discounts), which is valid for business analysis.

7. Exploratory Data Analysis (EDA)

Key insights derived from the dataset are summarized below.

Key Performance Indicators (KPIs)

- **Total Sales: 14.30 Million**
- **Total Profit: 3.83 Million**
- **Total Transactions: 5,000**

Sales by Category

- **Electronics** generated the highest sales, followed by **Fashion**.
- **Groceries** generated the lowest total sales among the three main categories.

Sales by Region

- The **East** region generated the highest revenue.
- Sales performance by region: East > West > North > South.

Payment Method Analysis

- **Cash** and **Credit Card** were the most frequently used payment methods.

Top 5 Stores by Sales

- **MegaMart New Michele** is the top-performing store by total sales.
- The other top stores include MegaMart Peckmouth, MegaMart Brianahaven, MegaMart Johnmouth, and MegaMart Jimenezborough

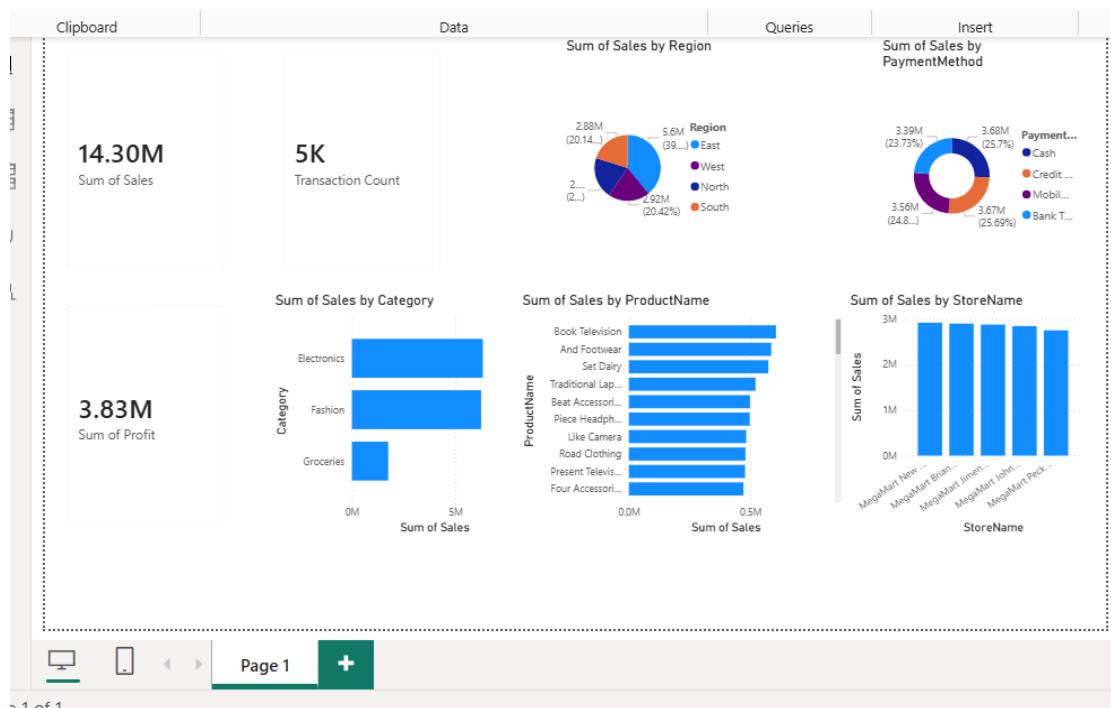
8. Dashboard Development

An interactive dashboard was created using **Power BI** based on the final_retail_dataset.xlsx.

Dashboard Components

- **KPI Cards:** Displaying Total Sales, Total Profit, and Total Transactions.
- **Bar Chart:** Sales by Product Category.
- **Bar Chart:** Sales by Geographic Region.
- **Bar Chart:** Top 5 Stores by Sales.
- **Bar/Pie Chart:** Sales Distribution by Payment Method.
- **Filters/Slicers:** To allow users to filter data by Date, Region, or Product Category.

The dashboard provides a visual and interactive way for stakeholders to monitor business performance, identify top-selling categories, and track regional trends to support data-driven decision-making.



9. Technologies Used

- **Programming Language:** Python
- **Data Processing:** Pandas
- **Development Environment:** Visual Studio Code
- **Data Storage:** Microsoft Excel (xlsx)
- **Data Visualization:** Microsoft Power BI

10. Conclusion

This project successfully implemented an end-to-end retail data engineering pipeline. Raw data from multiple source tables was ingested, cleaned, transformed, and validated using Python and Pandas. A final analytical dataset was generated, enabling exploratory analysis that revealed key insights, such as the dominance of the Electronics category and the East region in sales. Furthermore, a Power BI dashboard was developed to visualize these insights interactively. The project successfully demonstrates how data engineering practices can transform raw operational data into a valuable asset for business analytics and strategic planning.